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Review

2D-materials-based optoelectronic synapses for neuromorphic applications

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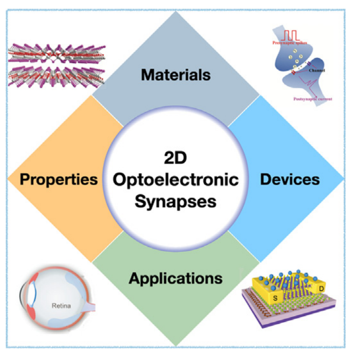
HIGHLIGHTS

- Two-dimensional (2D) optoelectronic synapses for artificial neural networks are systematically summarized.
- Recent progress of materials, synthesis methods, mechanisms and device configurations for 2D optoelectronic synapses is reviewed.
- Strategies for developing future neuromorphic vision and computing are summarized.
- Perspectives on and challenges in constructing 2D optoelectronic synapses for neuromorphic applications are discussed.

GRAPHICAL ABSTRACT



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ABSTRACT

Optoelectronic artificial synapses (OEASs) are essential for realizing artificial neural networks (ANNs) in next-generation information processing that has high transmission speed, high bandwidth, and low power consumption. Two-dimensional (2D) materials endowed with strong light-matter interactions and atomically thin dangling-bond-free surfaces are candidates for achieving versatile optoelectronics. Developing 2D OEASs for future neuromorphic applications is significant to break the bottleneck of von Neumann architecture and achieve future artificial intelligence systems. This review primarily focuses on recent developments in advanced 2D OEASs, discussing their working mechanism as well as potential applications. Common materials, device structures, and their synthesis and construction methods are also summarized. Finally, the prospects for future 2D OEASs from the perspectives of materials, performance, and applications are briefly described.

1. Introduction

The development of artificial intelligence has provided new challenges for processing and storing massive amounts of complex information. Artificial neuromorphic systems (ANSs) mimic the information processing of the human brain by achieving perception and reasoning tasks in one unit. Currently, artificial synapses (ASs) are essential elements of ANSs in the drive to achieve next-generation artificial intelligence systems. To mimic the synaptic behaviors of biological neural networks, ASs are endowed with multiple abilities, including short-term plasticity (STP), long-term plasticity (LTP), short-term depression (STD), long-term depression (LTD), paired-pulse

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facilitation (PPF), and spike-time-dependent plasticity (STDP), among others.

The synaptic weight regulation realized through changing resistance states is considered an essential part of a synaptic device. Although multifunctional neuromorphic devices composed of electronic synapses have been developed [1], ASs that use electrical stimuli as presynaptic input signals suffer from limited speed due to resistance-capacitance delay and the tradeoff between bandwidth and connection density [2,3]. Unlike electrically modulated synapses, OEASs perform synaptic functions by means of optical stimuli; they also offer the advantages of high transmission speed, high bandwidth, and low power consumption [4–6] and are compatible with other neural devices in advanced neural morphological networks [7,8]. Therefore, the development of new OEAS devices will contribute to the advancement of artificial intelligence systems.

Most of the reported OEASs simulate biological synaptic behaviors, and some devices reach an ultralow power consumption of < 0.1 aJ per synaptic event [9], but the majority still consume significant power compared with brain synapses (about 10 fJ per event) [10–12]. Plus, the devices' operating speeds are, for the most part, slower than biological synaptic signal transmission. It is thus challenging to realize optoelectronic synapses with low operating voltage, ultrafast operation speed, and low power consumption.

2D materials possess many intriguing and unique physical properties. In particular, 2D materials with strong light-matter interactions and atomically thin dangling-bond-free surfaces are potential candidates for realizing versatile OEASs [13,14]. Within the large family of 2D materials, their different bandgaps indicate a broad range of light absorption, from ultraviolet (UV) to infrared, which is beneficial for multi-wavelength modulation of synaptic behaviors [2]. Many recent ingenious studies of 2D optoelectronic devices have demonstrated their ability to perform artificial neuromorphic sensing and computation [15–17].

In this review, we mainly focus on OEASs that use 2D materials (Fig. 1). In the first section, we discuss 2D materials used in OEASs, including their properties and synthesis methods. In the following section, we categorize typical 2D OEASs into several groups according to their device configuration and explain their working principles. Applications based on these devices are then described, including neuromorphic computing and visual sensing. Finally, we summarize the review and offer our outlook on next-generation 2D OEASs.

2. Materials and synthesis of 2D optoelectronic synapses

2.1. Materials for 2D optoelectronic synapses

This section summarizes the representative 2D materials applied in optoelectronic synapses, including metallic, semiconducting, and insulating materials. A brief introduction to their properties is also given for deeper understanding.

Graphene has a unique Dirac-cone gapless energy band structure and high carrier mobility. Graphene's Fermi level can be modulated using electrical gating, and it exhibits p-type or n-type semiconducting behaviors under different gate voltages [18]. Although graphene has strong light-matter interaction in the range from ultraviolet to terahertz, the fairly low light absorption of monolayer graphene restricts its application in optoelectronics [19,20]. Perovskites can be used to broaden the absorption spectra of single-layer graphene [21]. For

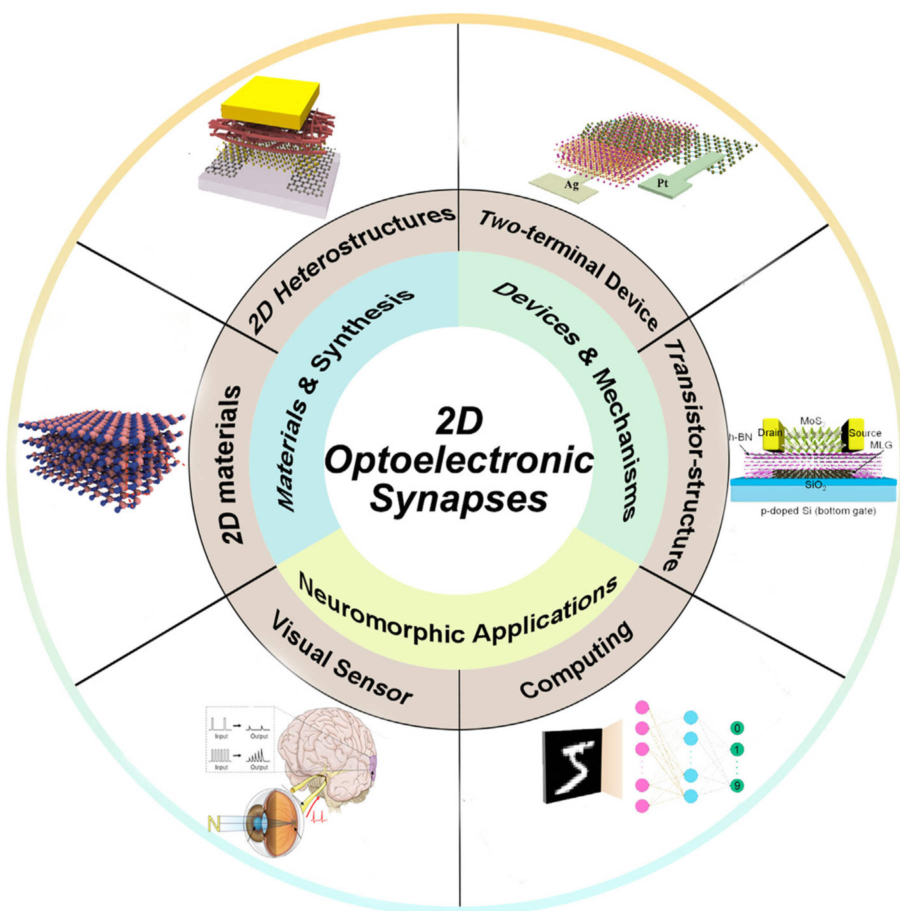

Fig. 1. An overview of 2D optoelectronic synapses used for neuromorphic visual sensors and computing applications. Reproduced with permission from Ref. [23]. Copyright 2021, Wiley-VCH. Reproduced with permission. from Ref. [62]. Copyright 2022, Wiley-VCH. Reproduced with permission. from Ref. [26]. Copyright 2022, Wiley-VCH. Reproduced with permission. from Ref. [95]. Copyright 2020, Springer Nature. Reproduced with permission. from Ref. [103]. Copyright 2021, American Chemical Society.

example, methylammonium lead bromide perovskite quantum dots (PQDs) were grown directly from a graphene lattice by using defect-mediated methods. Charge transfer occurred between $CH_3NH_3PbBr_3$ QDs and graphene flakes by π–π electron interactions. The as-synthesized phototransistors had a high responsivity of 1.4×10^8 A W^{-1} and a detectivity of 4.72×10^{15} Jones under visible light stimuli. When the PQD-decorated graphene was used in a phototransistor, it showed photonic synaptic behaviors such as STP and LTP due to photogenerated carriers and gate voltages.

2D hexagonal boron nitride (h-BN), also called white graphite, has a honeycomb hexagonal structure similar to graphene [22]. Due to its wide bandgap (~6 eV) — a result of the electronegativity difference between B and N atoms — h-BN is optically transparent and has low light absorption in the 250–900 nm wavelength range [23,24]. Insulating h-BN has a thickness-dependent dielectric property. Monolayer h-BN with a dangling-bond-free surface shows a static dielectric constant (3.3 eV) comparable to that of SiO_2, which enables it to function as a stable dielectric material tolerant of high temperatures [25]. For 2D heterostructures and optoelectronic transistors, h-BN is usually used for gate dielectric layers to suppress interface scattering and reduce fabrication-induced impurities [22,26].

MXenes are transition-metal carbides, nitrides, and carbonitrides synthesized by selective chemical etching of MAX phase precursors, which have layered structures with abundant surface terminations [27]. Most MXenes have high electrical conductivity and a tunable work function by altering the surface functional groups. Owing to their high hydrophilicity, MXene flakes can be well dispersed in water and fabricated into films via spin-coating or spray-coating, which is appropriate for the large-scale production of highly uniform transparent conductive films [28,29]. An optoelectronic transistor has been developed using partially oxidized MXene nanoflakes, with conductive MXene as floating gates and TiO_2 as tunneling layers [30]. The device showed UV-responsive optoelectronic synaptic behaviors and could be utilized in vision applications.

The formula of transition metal dichalcogenides (TMDs) is abbreviated to MX or MX_2, where M represents a transition metal (e.g., Ti, Mo, Ta, Nb, W, Re) and X is a chalcogen (S, Se, or Te) [31]. They have a finite bandgap of 1–2 eV and field-effect transistor (FET) mobility comparable to that of silicon [25]. MoS_2 is a representative TMD with two phases: semiconducting trigonal prismatic (2H) and metallic octahedral (1T) [32]. Monolayer MoS_2 has a direct bandgap that decreases as the number of layers increases, and a high electron mobility of 10–1000 $cm^2\ V^{-1}\ s^{-1}$. High-quality MoS_2 flakes with few structural defects are beneficial for high photoresponsivity and persistent photoconductivity [33]. For 2D materials similar to TMDs, the transition metals are replaced with post-transition metals (Ga, In, Sn, Tl, Pb, Bi) [25]. These also show layered 2D structures and typical semiconducting properties, and several representatives will be introduced in section 3.

Black phosphorus (BP) has semiconducting properties due to its puckered honeycomb structure, where phosphorus atoms are organized by sp^3 hybridization [19]. BP mainly shows p-type semiconducting properties and higher charge carrier mobility than TMDs [34,35]. The bandgap of 2D BP is also closely related to the thickness, but unlike most TMDs, it exhibits direct bandgap characteristics regardless of thickness [18]. A fully optical synapse based on mechanically exfoliated BP flakes had strong persistent photocurrents under UV stimulation [36]. The three-terminal device showed positive photoconductivity under 280 nm and 660 nm lights and negative photoconductivity under 365 nm, inducing excitatory and inhibitory postsynaptic current (EPSC and IPSC) behaviors. The unusual reduced photoconductivity of BP was explained by the influence of its surface adsorbates, which served as charge trap states and scattered the absorbed 365 nm light. By exploiting these behaviors, the device exhibited spike-duration-dependent plasticity (SDDP) and PPF. The device could be used to imitate Pavlovian conditioning and Hebbian synaptic learning rules. It also showed good flexibility when replacing Si substrates with polyethylene naphthalate (PEN) films.

2D perovskites with high mobility, strong optical absorption, and long carrier lifetime have attracted extensive attention owing to their huge potential for optoelectronic applications [37]. Their general formula, $(RNH_3)_2A_{n-1}B_nX_{3n+1}$ (n = 1, 2, 3, 4, …), consists of A positions (including methylamine (MA), formamidine (FA), Cs^+), B metal elements (e.g., Sn^{2+}, Pb^{2+}), and X halogen atoms (Cl, Br, I). When n = 1, the perovskites show 2D morphologies [38]. 2D perovskites are usually synthesized through a solution process, forming layered structures by inserting cations into 3D inorganic layers; this is easy to conduct and widely used in constructing optical synapses [39]. The common approaches include direct solution synthesis with hydrohalic acids, antisolvent methods using organic solvents, solvothermal synthesis, and solid-state grinding methods. In addition to their low-cost fabrication processes, 2D perovskites have high light absorption coefficients, high carrier mobility, and long diffusion lengths [40]. Moreover, their bandgap is easily tuned by modulating their chemical compositions and structures. The above properties offer huge potential for optoelectronic applications [41].

Some organic materials are also used in optoelectronic synapses for their low cost, strong scalability, high flexibility, and good biodegradability [42–44]. A light-stimulated synapse based on 2D 2,7-dioctyl [1] benzothieno [3,2-b] benzothiophene (C8-BTBT) films showed high photodetectivity and typical synaptic behaviors [45]. 2D metal-organic frameworks (MOFs) are also excellent charge-trapping materials and can be used as synaptic devices. A three-terminal device employing Zn-TCPP (TCPP: tetrakis (4-carboxyphenyl)porphyrin) demonstrated synaptic behaviors and learning functions [46]. Additionally, some MOFs exhibit reliable broadband photoresponsivity that is beneficial for OEASs [47].

In spite of their superb electrical and optoelectronic properties, single 2D materials still have various problems. Being ultrathin, they always suffer from low carrier mobility and limited light absorption. For instance, perovskites with broadband light absorption usually will be combined with TMDs to yield more photogenerated charge carriers. Therefore, coupling 2D materials with atomically sharp interfaces is beneficial for optimizing heterostructure band alignments and optical transactions, which help to modulate charge carrier transport and the photogating effect.

2.2. Synthesis and fabrication techniques for 2D optoelectronic synapses

The synthesis and fabrication procedures for 2D materials and synapses are essential to guarantee the quality and optoelectronic performance of the resulting electronics, and thus the choice of fabrication method largely depends on the material types and target applications. For OEASs, the most common synthesis methods include mechanical exfoliation, liquid phase exfoliation, and chemical vapor deposition (CVD).

Liquid-phase exfoliation is a top-down process to facilitate compact thin films made of a wide range of 2D flakes. The interlayer van der Waals force is weak, so this method produces monolayers at large scale by exploiting solution-processed intercalation and exfoliation. To construct synaptic devices, the as-fabricated 2D flakes are usually further assembled into films through spin-coating, spray-coating, or dip-coating. However, the fabricated nanosheets have smaller size, more defects, and less-controllable thickness than nanosheets synthesized by CVD or mechanical exfoliation. Liquid-phase exfoliation leads to low carrier mobility in 2D semiconductors, low electrical conductivity in 2D electrodes, and low dielectric constants for 2D insulators [35].

Mechanical exfoliation is the most popular method used in the lab to produce 2D flakes with clean surfaces that are free of dangling bonds. However, it is impossible to produce large-scale 2D films by this method, since precise control over size and thickness is not only limited by the bulk precursor but also unachievable in large quantities.

CVD is considered an effective strategy to grow high-quality, large-area 2D materials and heterostructures [48]. During CVD, precursors are transported with the carrier gases, and the 2D products of chemical reaction deposit on the substrate to certain degrees. This method provides 2D materials with controlled thickness, nucleation rate, flake morphology, and surface roughness. However, CVD-grown 2D materials have many inherent defects [49], which strongly affect the charge trapping/de-trapping process.

Constructing 2D heterostructures for optimized performance is more convenient than modifying single materials by doping or decorating [50]. The absence of dangling bonds from the surface of 2D materials dispels concerns about lattice mismatch and synthesis incompatibility. The most common and straightforward way to produce a 2D heterostructure is to use a top-down strategy by mechanically exfoliating bulk materials, and then transferring 2D flakes to the desired location by a polymer-assisted process viewed through an optical microscope. The transfer process can be either dry or wet. During a dry-transfer process, a sacrificial thermal release polymer layer such as polydimethylsiloxane (PDMS) is used to peel the monolayers from 2D bulk materials. During a wet-transfer process, polymethyl methacrylate (PMMA) has the same function as PDMS, except that PMMA films need solidification and should be removed in acetone or acetone vapor [19]. But the difficulty of completely removing the polymer residue introduces interfaces and contamination that impair carrier transport [51], so for 2D heterostructures, thermal annealing is usually conducted as the last step to release the induced trap states and improve the interfacial quality. An additional challenge is that during transfer, any stretching or bending of the supported polymer will result in wrinkles on the 2D material. In contrast, a dry-transfer process enables a cleaner surface and less contamination and therefore is better for constructing heterostructure optoelectronics.

Compared with thin films composed of 0D and 1D nanomaterials, the large aspect ratio of 2D films means they have fewer in-plane grain boundaries, ensuring better optoelectronic performance. In addition, the van der Waals heterostructure composed of 2D flakes yields quite small transport barriers for charge carriers to migrate across neighboring flakes [52]. Moreover, 2D materials well dispersed in solvents can form stable colloidal inks, offering a potential strategy for developing standardized production via related printing techniques and solution processes [53]. This method is suitable for preparing flexible electronics via cost-effective manufacturing on a large scale.

3. Device structures and operating mechanisms of 2D optoelectronic synapses

3.1. Fundamentals of 2D optoelectronic synapses

The main mechanisms by which optoelectronic devices transform optical stimuli into electrical signals are photoconduction, photogating, and the photovoltaic effect. Many OEASs work via the persistent photoconductivity of 2D semiconductors, meaning the photocurrents last even when light stimuli are removed [54]. The persistent photoconductivity is mainly due to defect-mediated carrier trapping, phase changes, ion migration, or energy barriers that suppress charge-carrier recombination [55]. This photoelectric effect is usually applied to explain STDP behaviors, and it is the principle behind both short-term and long-term plasticity.

For OEASs, device resistance is considered synaptic weight. It is essential to control the transport and recombination of photogenerated carriers so as to precisely modulate the device resistance. The charge-carrier trapping/de-trapping process has an important effect on the photogenerated carriers transport. When one type of photocarrier (i.e., electrons or holes) is trapped by intrinsic or extrinsic trap states, the other type will circulate within the circuits, and its lifetime will be prolonged, so the photocarriers will have a reduced recombination rate [31]. When the trapped charge carriers are released because of thermal fluctuation, electron–hole recombination will decrease the postsynaptic currents. In addition, the built-in electrical field induced by 2D heterostructures can separate the generated electron–hole pairs and suppress the recombination process. The following section will describe some typical 2D OEASs.

3.2. Device structures and operating mechanisms

According to their structure, OEASs can be classified as two-terminal or transistor-structure devices. In this review, two-terminal devices are mainly synapses composed of photosensitive materials, and they have a simple structure and low power consumption. Memristors are typical two-terminal devices located at the intersections forming artificial neural networks [56]. Using two-terminal memristors as synapses offers the high connectivity and high processor density necessary to mimic efficient biological systems [57]. 2D conductors, semiconductors, and insulators, such as graphene, MoS_2, and BN, have been used in memristors because of their atom-thick layered structures, unique optoelectronic properties, and the superb mechanical flexibility of the resulting materials. Two-terminal OEASs that use 2D materials can be classified into vertical devices, lateral devices, and heterostructures, according to their device configuration and working mechanism.

Three-terminal or multi-terminal phototransistors exploit the photogating and electrical gate effects, whereby output currents can be modulated optically and electrically. For 2D phototransistors, the photogating effects result from trap states [58]. Therefore, the synaptic behaviors are closely related to the interface between the semiconductor channel and dielectric layer, where suitable charge trapping is influenced by device configuration [59].

Here, we separate 2D optical synapses into two-terminal and transistor-structure types (Table 1) and clearly explain the specific mechanism of each type.

3.2.1. Two-terminal 2D optoelectronic synapses

In lateral devices, the charge carriers flow along the in-plane direction, so defects in the 2D channel affect the photoconductive behaviors. $PtSe_{2-x}$ bi-directional photonic synapses have been obtained via controlled selenization engineering, yielding partially selenized films that include $PtSe_2$ and metallic Pt (Figs. 2a–d) [60]. By modulating the $PtSe_{2-x}$ film's composition, the photoconductivity of the device was also changed, falling between the negative photoconductivity of Pt due to its bolometric effect and the positive conductivity of $PtSe_2$ resulting from photogenerated charge carriers. Furthermore, the difference between the light absorptions of Pt and $PtSe_2$ in company with the various compositions also contributed to the photoconductivity transition. Thus, controlling the wavelength of light stimuli was feasible to realize fully optical $PtSe_{2-x}$ synapses. The $PtSe_{1.66}$ device possessed EPSC and IPSC characteristics, and its PPF index reached 182%.

Although two-terminal vertical devices suffer from large power consumption, their ultrathin nature enables the downscaling of integrated circuits. For chip-level integration, vertical integration may be the most practical way to increase chip density. Ternary 2D BiOI has a bandgap of 1.6–1.9 eV, and its structure is beneficial for charge separation and transfer, demonstrating promising optoelectronic properties [61]. The reported BiOI-based optoelectronic memristors (Figs. 2e–h) [62] were used to construct a two-terminal vertical structure with BiOI sandwiched between Ag and Pt electrodes. After electroforming, the device showed a typical bipolar resistive switching behavior, and the SET and RESET voltages were quite low ($\pm$0.05 V), indicating low power consumption. The resistive switching occurred by typical electrochemical metallization, and the conductive channel was formed and reversed owing to Ag^+ migration. The device had good retention and endurance. Under electrical spikes, the synaptic memristor showed STP and LTP. When Pt electrodes were replaced with graphene, a 400 nm light pulse stimulated the photoresponse current of BiOI, and STP and LTP were also achieved due to the generation and recombination of photo-induced carriers. Under different sequences of 400 nm light, the

Table 1
Representative 2D optoelectronic synapses and their key parameters.

Active materials	Device structure	Wavelength (nm)	Synaptic functions	PPF max index/%	Operation speed/ min pulse width	Positive & negative bi-directional photoresponse	Reading voltage/V	Power/energy consumption per event	Reference
$PtSe_{2-x}$	two-terminal	405–5000	EPSC, IPSC, PPF	182	200 ms	Y	1	–	[60]
BiOI	two-terminal, vertical	400	PPF, LTP, LTD	~90	2 s	N	0.2	10 pW	[62]
$MoSe_2/Bi_2Se_3$	two-terminal	790	EPSC, IPSC, PPF, PPD LTP, STP	15 (PPD)	0.1 ms	N	0.5	100 pJ	[64]
perylene/GO	two-terminal	365–1550	EPSC, PPF, STP, LTP, LTD, SIDP, SNDP	214	1 s	N	1	10^{-9} W	[102]
MoSSe	transistor	450	PPF, STP, STD, LTP, LTD, SNDP	166	50 ms	N	–	–	[70]
InSe	transistor	405, 520, 650	PPF, STP, LTP	–	0.1 s	N	0.5	–	[74]
MoS_2	transistor	200, 450, 1000, 2000	STP, LTP	–	50 ms	N	1	–	[86]
black phosphorus	transistor	280, 365, 660	EPSC, IPSC, SDDP, STDP, STM, LTM	285	50 ms	Y	0.01	3.5 pJ (optical)	[36]
α-In_2Se_3	transistor	900	STP, LTP	>1000	0.5 s	N	0.3	–	[84]
α-In_2Se_3	transistor	655	EPSC, IPSC, PPF, PPD, LTP, LTD	~40	50 ms	N	−0.1	–	[71]
2D MOF	transistor	420	EPSC, PPF, STP, LTP	125	–	N	1	–	[73]
$(PEA)_2SnI_4$	transistor	460–470, 520–530, 620–630	PPF, STP, LTP	129.7	20 ms	N	2	–	[72]
$MoS_2/BaTiO_3$	transistor	450, 532, 650	LTP, LTD	–	100 ms	N	–	1.8 pJ	[85]
Bi_2O_2Se/Gr	transistor	365, 635	EPSC, IPSC, PPF, LTP, LTD, SRDP	~120 (EPSC), ~190(IPSC)	100 ms	Y	0.5	–	[68]
graphene/MoS_2	multi-terminal	525	LTD	–	50 ms	N	1	–	[76]
$(PEA)_2SnI_4$/Y6	transistor	UV-NIR	EPSC, IPSC, PPF, STM, LTM	160	20 ms	Y	±40	~0.925 nJ	[94]
pyrenyl graphdiyne/ graphene/PbS	transistor	450，980	EPSC, IPSC, PPF, LTP, LTD	~180 (EPSC), ~190(IPSC)	20 ms	Y	0.01	0.02 pJ (980 nm), 0.1 pJ (450 nm)	[103]
α-In_2Se_3/GaSe	transistor	450	PPF, LTP, LTD	~21	1 s	N	0.5	–	[82]
$ReSe_2$/h-BN/graphene	transistor	405	EPSC, IPSC, SNDP, STDP	–	30 ms	N	5	–	[13]
h-BN/MoS_2	transistor	470	PPF, STP, LTP, LTD	~270	50 ms	N	0.01	2.52 fJ	[26]
2D polymer	transistor	405	EPSC, PPF, STM, LTM	165	500 ms	N	−0.1	0.29 pJ	[75]
$MAPbBr_3$/graphene	transistor	430–440	PPF, PPD, STP, LTP, LTD	~165	5 s	N	0.5	36.75 pJ	[21]
$CsPbBr_3/MoS_2$	transistor	405	EPSC, STP, LTP, STM, LTM	–	10 ms	N	0.1	42.93 nJ	[40]
MoS_2/P(VDF- TrFE)	transistor	405	EPSC, IPSC, LTP, STM, LTM	–	2 s	N	–	–	[80]
$WS_2/PbZr_{0.2}Ti_{0.8}O_3$	transistor	532	STP, LTP, LTD, STM, LTM	–	100 ms	N	0.1	–	[79]

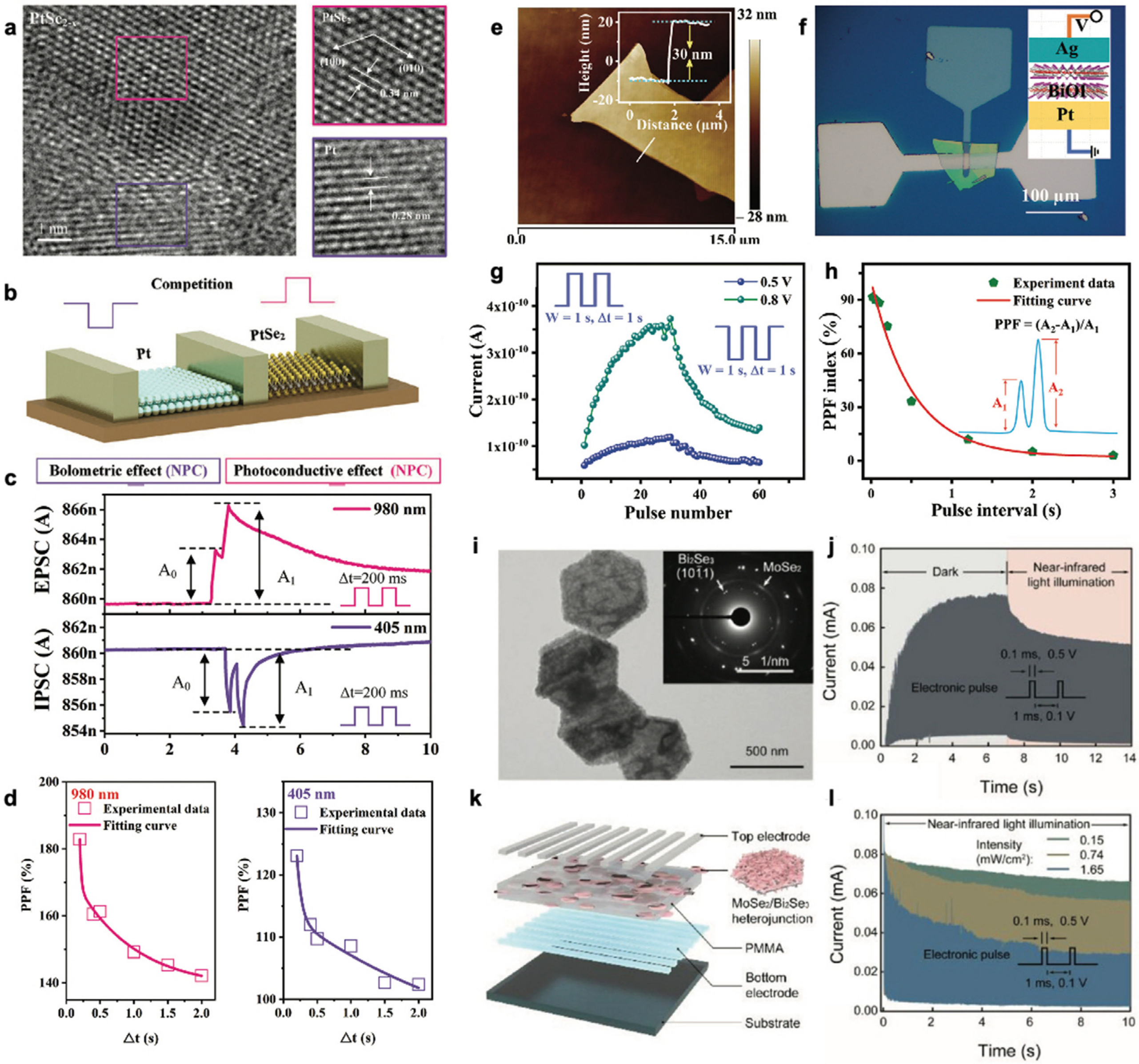

Fig. 2. Introduction of two-terminal optoelectronic synapses. (a) High-resolution transmission electron microscope (HRTEM) images of $PtSe_{2-x}$ films, indicating distinctions in the transformation process; red region, $PtSe_2$, violet region Pt. (b) Schematic illustration of the $PtSe_{2-x}$ device. (c) EPSCs and IPSCs triggered by 405 nm pulses. (d) Relationship between PPF index for $PtSe_{1.66}$ device and time interval Δt. (a–d) Reproduced with permission. from Ref. [60]. Copyright 2022, Wiley-VCH. (e) AFM image of exfoliated BiOI nanosheets. Inset: height profile. (f) Optical microscope image of Ag/BiOI/Pt memristor. (g) LTP and LTD curves with pulse amplitudes of 0.5 and 0.8 V. (h) PPF index of the device. Inset: change in postsynaptic current induced by a pair of input signals. (e–h) Reproduced with permission. from Ref. [62]. Copyright 2022, Wiley-VCH. (i) TEM image of $MoSe_2/Bi_2Se_3$ nanosheets. Inset: SAED pattern of the sample. (j) Device currents with NIR light on the memory device, turning PPF into a paired-pulse depression (PPD) effect. (790 nm, 1 mW cm^{-2}). (k) Schematic of $MoSe_2/Bi_2Se_3$ RRAM device. (l) Current responses under consecutive electronic pulses with different light stimuli. (i–l) Reproduced with permission. from Ref. [64]. Copyright 2019, Wiley-VCH.

currents of the BiOI memristor exhibited typical learning-experience behaviors.

Using heterostructures is a common strategy to greatly improve synaptic characteristics. 2D topological insulators have a narrow bandgap and strong absorption of near-infrared (NIR) light. Among them, Bi_2Se_3 demonstrates optical broadband absorption and polarization-sensitive photoresponse [63]. A $MoSe_2/Bi_2Se_3$ heterostructure has been fabricated using all-solution processing (Figs. 2i–l) [64]. Polymethyl methacrylate (PMMA) was added into the active layer to passivate the trap states in the heterostructure. NIR was used to stimulate electron–hole pairs and hinder the formation of conductive filaments, and thereby control the device's conductivity and synaptic plasticity. When the heterostructures were arranged as RRAM arrays, they could memorize visual information.

3.2.2. Transistor-structure 2D optoelectronic synapses

This section introduces several new transistors that use 2D materials for OEASs. Oxychalcogenide Bi_2O_2Se with a bandgap of 0.8 eV and a small effective electron mass shows high chemical stability in air when synthesized using oxidation CVD [65]. Unlike 2D materials bonded with van der Waals force, Bi_2O_2Se layers are bonded with weak electrostatic interactions. 2D optoelectronics using Bi_2O_2Se have attracted much interest because of the latter's high electron mobility and broadband photoresponse properties, and because Bi_2O_2Se FETs show a low subthreshold swing [66]. Bi_2O_2Se infrared photodetectors exhibit high quantum efficiency [67], and Bi_2O_2Se/graphene heterostructures can mimic biological neuroplasticity (Figs. 3a–d) [68]. Bi_2O_2Se and graphene were synthesized by a low-pressure CVD process, then the device was fabricated by dry

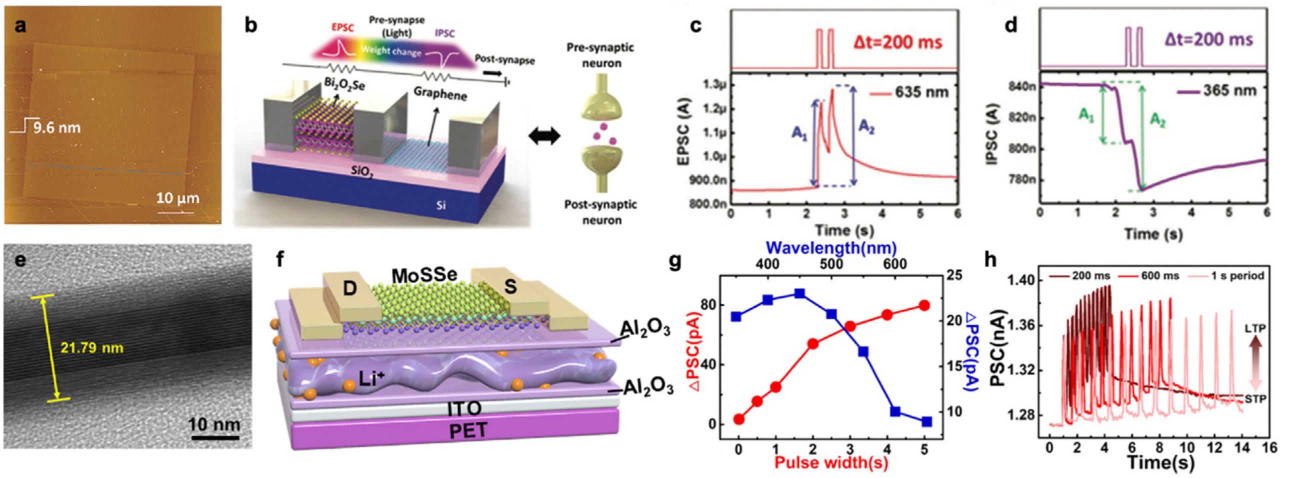

Fig. 3. Description of two-terminal synaptic devices. (a) AFM image of a single Bi_2O_2Se nanoplate. (b) Schematic illustration of a Bi_2O_2Se/graphene optoelectronic synapse. (c) EPSCs and IPSCs induced by the pair of presynaptic optical pulses. (d) Dependence of PPF index on Δ*t*. (a–d) Reproduced with permission. from Ref. [68]. Copyright 2020, Wiley-VCH. (e) HRTEM of 2D MoSSe. (f) Scheme of 2D Janus MoSSe electronic device. (g) Postsynaptic current change (ΔPSC) of the device. (h) STP to LTP conversion by applying different periods of light pulses. (e–h) Reproduced with permission. from Ref. [70]. Copyright 2022, American Chemical Society.

transfer. Under 635 nm illumination, the Bi_2O_2Se photodetector showed positive photoconductivity, whereas under 365 nm illumination, the graphene showed negative photoconductivity. The authors inferred that the higher photocurrent of Bi_2O_2Se was caused by both photoconduction and the bolometric effect, and the reduced currents of graphene were caused by light-induced oxygen adsorption/desorption. By exploiting the opposing photoresponses of the heterostructures, the authors modulated the wavelength and intensity of light stimuli to trigger the resistance change as synaptic weight. By combining the EPSCs of excitatory Bi_2O_2Se and the IPSCs of inhibitory graphene, a tandem OEAS was constructed using two two-terminal lateral devices, and the resulting dual-mode photoresponse realized fully optical regulation of the synapse without employing a gate controller. Under stimulation from multiple light spikes of varying frequencies, the device showed short-term and long-term memory, as well as LTP and LTD.

Monolayer MoSSe breaks out-of-plane symmetry and has superb physical and chemical properties [69]. A transistor that used 2D Janus MoSSe could be modulated by light and electrons or ions (Figs. 3e–h) [70]. The device used mechanically exfoliated MoSSe as the channel and Al_2O_3 as the trapping layer, while a polyimide (PI) solution containing lithium ions was fabricated as an electric double layer to lower the gate voltage. Electrical modulation and Li^+ migration helped achieve synaptic plasticity, including EPSC/IPSC, STP, and LTP. When light illuminated the device, photogenerated electrons decreased the resistance of the oxide layer and increased the channel current. As the interval between stimuli increased from 100 ms to 3 s, the PPF index was lowered from 166% to 104%. The device was tested as a flexible artificial retina. With the assistance of a gate-voltage pulse, the device could mimic the light-adaptation process of retina cells. The authors then designed an artificial neural network to process and recognize image information, and using the MoSSe device increased the recognition rate from 77.6% to 83.3%.

A ferroelectric α-In_2Se_3 OEAS has been developed [71]. The α-In_2Se_3 flakes used in the device had a rhombohedral (*R*3m) structure, and the in-plane and out-of-plane polarizations showed dipole locking effects. When positive and negative voltage pulses were applied to the device, the postsynaptic currents revealed typical STP and STD effects, respectively. As the negative voltage amplitude increased, the currents did not recover to the original states, and STD changed to LTD. The reduced conductivity resulted from charge traps and ferroelectric switching processes. The PSC increased when a 655 nm light pulse was applied to stimulate the α-In_2Se_3 device, then decayed to the initial state because of fast photoconduction decay and slow trap decay. By coupling electric and optoelectronic plasticity, the inhibition effect and relaxation time could be decreased through applying light stimulation and negative back-gate voltages. The relationship between ferroelectric and optoelectronic properties indicated that (i) strong enough voltage pulses would eliminate light-induced fading memory and (ii) the ferroelectric conductivity changed depending on the time interval between electrical and light pulses. The authors designed a mixed-input reservoir strategy to increase the accuracy of recognizing handwritten digits. Due to the mixed electrical and optoelectronic mechanism and the nonlinear transformation of inputs, the recognition accuracy was increased to 98.6%. Moreover, the relaxation times of the device were tunable, so it could be integrated with reservoir computing systems that work at multiple timescales, and a parallel-reservoir computing system was used to perform multiscale signal processing.

A floating-gate device has been developed, with $ReSe_2$ as the channel and photoactive layer and graphene as the charge storage layer (Figs. 4a–f) [13]. The device was operated by a photoactive electro-controlled mechanism and created a controlled memory state. The postsynaptic current response in the device could be controlled by optical pulses, and it realized optically modulated EPSC and IPSC, as well as SNDP. Owing to the device's inequivalent Schottky barrier heights, IPSC had a faster training speed than EPSC, as electrons transported more easily than holes. The device emulated Hebbian learning rules, and a spiking neural network was built to show the potential for image recognition.

Elsewhere, a 2D $(PEA)_2SnI_4$ perovskite OEAS was demonstrated (Figs. 4g–i) [72]. Under light stimuli, the transistor showed typical EPSC effects, and STP behavior could be transformed to LTP by changing the light spikes' frequencies. The slow photogenerated electron–hole recombination process of $(PEA)_2SnI_4$ led to short-term memory and learning behaviors. The synaptic currents of the $(PEA)_2SnI_4$ device changed according to the light spectrum, with white and blue stimuli generating the largest PSCs. Additionally, the chemical composition of $(PEA)_2SnI_4$ could be easily tuned, and these changes affected the optoelectronic synaptic functions. By increasing the number of light spikes, STP could be converted to LTP. The trapping mechanism was applied to explain these characteristics, and Sn vacancies were assumed to catch photogenerated electrons and cause the photogating effects.

A synapse that uses a 2D MOF has been achieved. The active layer was Cu-THPP (THPP = 5,10,15,20-Tetrakis (4-hydroxyphenyl)-21H, 23H-porphine) [73]. The synthesized Cu-derived MOF had semiconducting properties, good chemical stability in air, and quite a large specific

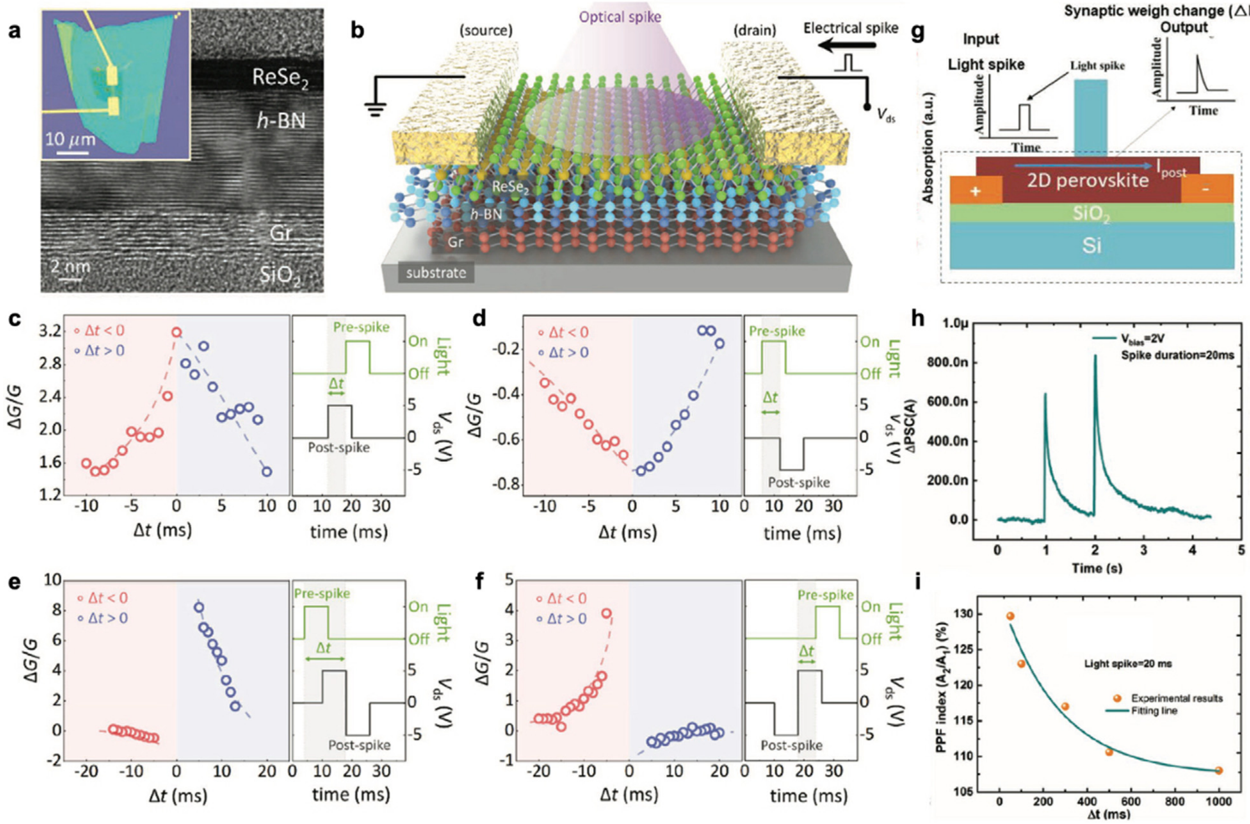

Fig. 4. Description of heterostructure synaptic devices. (a) TEM image of $ReSe_2$/h-BN/Gr heterostructures. Inset: optical image of $ReSe_2$/h-BN/Gr heterostructure. (b) Scheme of $ReSe_2$/hBN/Gr heterostructure. Relationship between ΔG/G (the conductance change between the pre- and post-synaptic simulations) and the time interval Δt, emulating the STDP behavior of (c) symmetric Hebbian, (d) symmetric anti-Hebbian, (e) asymmetric Hebbian, and (f) asymmetric anti-Hebbian learning rules. (a–f) Reproduced with permission. from Ref. [13]. Copyright 2021, Wiley-VCH. (g) Illustration of $(PEA)_2SnI_4$ devices. The insets are examples of the input light pulses and the output PSC. (h) ΔPSC triggered by a pair of light spikes. (i) PPF index versus light interval Δt between light pulses. PPF index decreases exponentially as Δt increases. (g–i) Reproduced with permission. from Ref. [72]. Copyright 2019, Wiley-VCH.

surface area. The monolayer Cu-THPP film showed strong absorption of light at 420 and 550 nm. Applying 420 nm light as the input signal, the OEAS showed typical photoconductivity and exhibited STP behaviors. When continuous light stimuli were applied with various frequencies and spike numbers, the STP would convert to LTP, and the device showed great potential for memory improvement. The maximum PPF index of the device was 125% with a 50 ms interval between stimuli. MOFs have broadened the range of materials that can be used in OEASs.

In 2020, a 2D phototransistor that uses InSe was reported as a synaptic device [74]. The researchers inferred that carrier trapping dominated by the trap states of SiO_2 substrates was the origin of persistent photocurrents in the InSe device. As the frequency of light stimuli was raised, STP changed to LTP, and higher light intensity and light pulse width increased the output photocurrents. Moreover, the PPF index could be tuned by adjusting the applied gate voltages, and the relationship between PPF and time intervals could be transformed according to the applied gate bias.

2D imine polymer has been used as the charge-trapping layer of OEASs [75]. The polymer was synthesized by Schiff base reaction using benzene-1,3,5-tricarbaldehyde (BTA) and p-phenylenediamine (PDA). Photogenerated carrier migration was modulated by light stimulation and gate voltages. The PPF index of the device was 163%, and its power consumption was ~0.29 pJ per spike.

By integrating different modes of synapse plasticity, synaptic weight can be updated to achieve multifunctional artificial intelligence. Graphene/MoS_2 heterostructure transistors with a triboelectric nanogenerator (TENG) were able to collect both biomechanical and visual information [76]. Coupling with the TENG modulated the photogenerated electrons' transfer from MoS_2 to graphene and thereby change graphene's resistance. By altering the mechanical displacement of the TENG at the device's bottom, as well as the light illumination on the heterostructure, the postsynaptic current was activated and then relaxed; this response was similar to LTD in a biological synapse. Changing the intensity and duration of light illumination modulated the charge carrier migration and affected the synaptic inhibitory process. The synergistic effect of mechanical and visual information offered the opportunity to construct a multilayer perception-based artificial neural network that contained 28×28 input neurons, 100 hidden neurons, and 10 output neurons. A supervised learning function that exploited mechano-photonic plasticization helped to achieve image recognition with 92% accuracy. This mechano-photonic AS led to mixed-modal plasticity and the emulation of neuromorphic computation.

Similar to two-terminal devices, heterostructures also help enhance photocarrier migration and synaptic behaviors [77]. Each layer of a 2D material is covalently bonded, and the layers are stacked by weak van der Waals force, leading to a dangling-bond-free surface and precisely controlled charge carrier migration. 2D heterostructures are no longer constrained by lattice matching and synthesis compatibility.

A van der Waals 0D-2D heterostructure phototransistor using $CsPbBr_3$ quantum dots was able to increase the light-absorption capability of 2D MoS_2, which demonstrated high carrier mobility [40]. The heterojunction had a Type I band alignment structure, and the photo-induced electron flows and trapped holes formed photogating effects, which also affected the long-term synaptic behaviors. Elsewhere, a floating-gate transistor employing a MoS_2/h-BN/graphene van der Waals heterostructure has been reported [26]. In the device, photogenerated electrons from MoS_2 tunneled through the h-BN layer and migrated between multilayer graphene and MoS_2. The device emulated optoelectronic plasticity and had an ultralow power consumption of 2.52 fJ per spike. A two-layer neural network was constructed for handwriting recognition and achieved a recognition accuracy of 92.2%.

Ferroelectricity offers a new way of modulating the charge-transmission process. For 2D synapses, a polarized electric field can be provided by $BaTiO_3$ oxides [78], ferroelectric $PbZr_{0.2}Ti_{0.8}O_3$ (PZT) films [79], organic poly (vinylidenefluoride-co-trifluoroethylene) [P(VDF-TrFE)] [80], or 2D ferroelectric flakes. Integration with highly incompatible materials can be achieved with 2D ferroelectric materials [81]. For example, 2D ferroelectric α-In_2Se_3 with a bandgap of 1.4 eV has been combined with p-type GaSe as the active layer to form an OEAS device (Figs. 5a–c) [82]. The heterostructure was fabricated by mechanical exfoliation and dry transfer. When the three-terminal device was exposed to light stimulation, photogenerated carriers increased the channel currents, which then decreased when the light was removed. Pavlovian conditioning was mimicked by combining light pulses that induced large PSCs, and smaller currents with electrical pulses. According to the authors, α-In_2Se_3 exhibited reversible spontaneous polarization in both in-plane and out-of-plane orientations, and the strong polarization effect could be used to store data. The authors formed a crossbar and constructed an ANN by using ferroelectric transistors as synaptic devices, and accuracy at identifying handwritten digits reached 90% after 1000 training epochs.

The ferroelectric domain can be engineered electrically and optically for some materials, which have potential uses as non-volatile optoelectronic memories [83]. α-In_2Se_3 flakes have demonstrated broadband optoelectronic memories (Figs. 5d–f) [84]. A two-terminal lateral device showed non-volatile photoresponse properties under various domain states arising from changes in the ferroelectric ordering. These traits were exploited to achieve ferroelectric optoelectronic memory, with the optical-writing and electrical-erasing ratio reaching 10^4. The device was able to mimic the STP and LTP features of synaptic plasticity under light pulse stimuli.

A heterostructure of 2D MoS_2 and ferroelectric $BaTiO_3$ was shown to act as non-volatile memory and a neuromorphic visual sensor [85]. The ferroelectricity of $BaTiO_3$ was modulated by light stimuli, and the ferroelectric polarization switching depended on the light dosage, owing to interactions between the photo-induced carriers of MoS_2 and polarized charges of $BaTiO_3$ (Figs. 5g and h). Based on the above mechanisms, a neuromorphic visual sensing system was constructed, and the recognition accuracy for handwritten digits was enhanced to 91%.

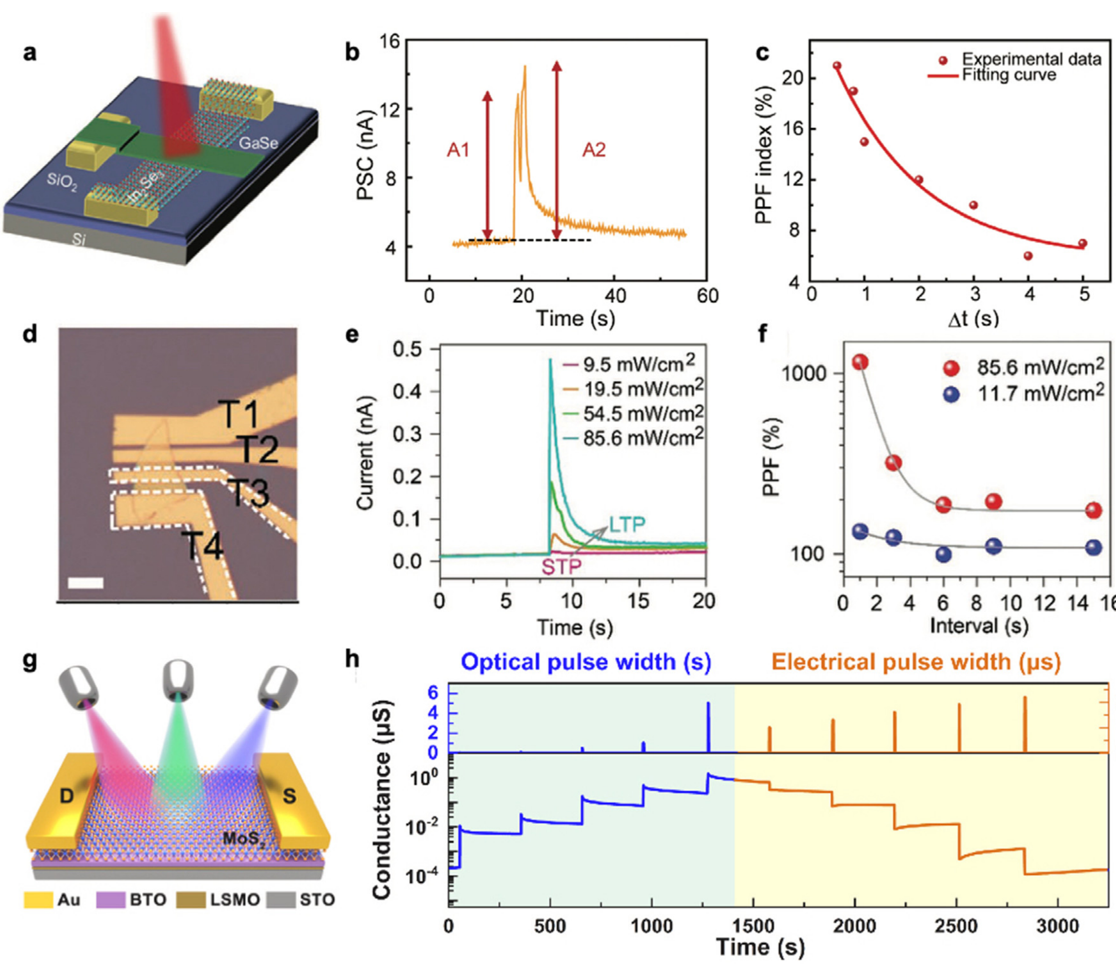

Fig. 5. Illustration of transistor-structure ferroelectric 2D optoelectronic synapses. (a) Structure of multifunctional α-In_2Se_3 device. (b) PSCs triggered by a pair of light pulses (808 nm, 7.8 mW cm^{-2}, width 1 s, $\Delta t = 0.5$ s). (c) PPF index as a function of light pulse interval Δt. (a–c) Reproduced with permission. from Ref. [82]. Copyright 2022, Wiley-VCH. (d) Optical image of α-In_2Se_3 device. Scale bar: 5 μm. (e) Time-resolved currents under different light power densities, and transition from STP to LTP. (f) Emulation of PPF under two identical light pulses separated by different intervals. (d–f) Reproduced with permission. from Ref. [84]. Copyright 2020, Wiley-VCH. (g) Scheme of MoS_2/$BaTiO_3$ synapses stimulated by three light pulses (450, 532, and 650 nm). (h) Non-volatile multilevel conductance switching under light stimuli ($\Delta t = 50$, 100, 500, 1000, or 5000 ms) and electrical excitations ($\Delta t = 1$, 1.3, 1.6, 1.9, or 2.2 μs). (g, h) Reproduced with permission. from Ref. [85]. Copyright 2021, Elsevier.

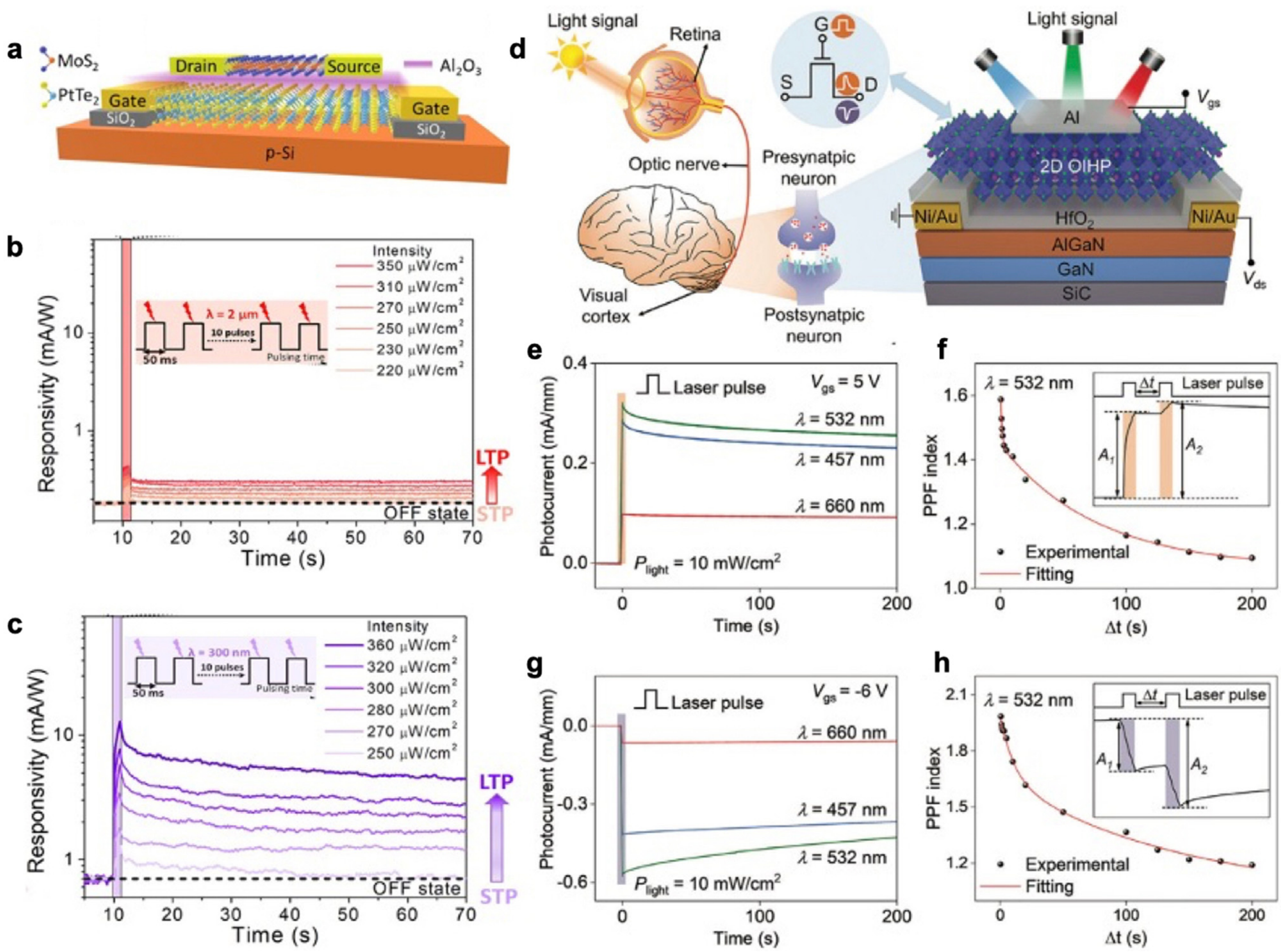

Fig. 6. 2D materials applied as gates of FETs. (a) Schematic diagram of $PtTe_2$/Si-gated optoelectronic synapse. Transition from STP to LTP as a function of light intensity under illumination of wavelengths (b) 2.0 μm and (c) 300 nm. Reproduced with permission. from Ref. [86]. Copyright 2022, American Chemical Society. (d) Scheme of human visual system (left) and $(PEA)_2PbI_4$-gated visual sensor (right). (e) Photocurrents induced by 457/532/660 nm laser pulses at $V_{gs} = 5$ V. (f) PPF index triggered by 532 nm successive pulses at $V_{gs} = 5$ V. (g) Photocurrents at $V_{gs} = -6$ V. (h) PPF index triggered by 532 nm successive pulses at $V_{gs} = -6$ V. Reproduced with permission. from Ref. [87]. Copyright 2022, Wiley-VCH.

2D materials can also be applied as gates of FETs to increase their optical absorption. Using $PtTe_2$/Si as gate electrodes increased the infrared absorption of MoS_2 FETs (Figs. 6a–c) [86]. 2D $PtTe_2$ and MoS_2 were synthesized by CVD, and the MoS_2 was transferred to the top of the device. The device was sensitive to multiple wavelengths, and its conductance could be modified by photogenerated carriers and the photogating effect. The device also emulated STP and LTP. Moreover, an ANN that used this device achieved 80% recognition accuracy of multicolored written patterns reflected by infrared, near-infrared, visible, and UV light.

Very recently, a bi-directional AlGaN/GaN photonic synapse for visual applications was realized using 2D perovskite $(PEA)_2PbI_4$ as the gate (Figs. 6d–h) [87]. Increased light intensity markedly lowered the activation energy of ion transport in the perovskite gates, and measurable photoresponse was therefore detected in the range of ultraviolet and visible light. In addition, the light stimuli enhanced the field-effect modulation and endowed the device with both positive and negative photoconductivity. The device emulated EPSC, IPSC, and PPF behaviors. When organized in arrays, the synapses constituted a neuromorphic visual system that achieved 100% recognition of color images.

In 2018, modified 2D graphene oxide (GO) nanosheets were used as charge-trapping layers. The nanosheets were modified by being embedded with long alkyl chains, then were used in a photonic synapse device that included an ion-gel dielectric layer and an indium-gallium-zinc oxide (IGZO) photosensitive semiconductor [88]. The charge de-trapping process was improved by light pulse stimulation, inducing increased synaptic weight tunability and enhanced recognition accuracy of digit patterns.

4. Neuromorphic applications

4.1. Neuromorphic visual sensors

Distinct from conventional artificial visual systems consisting of image sensors, memory, and processing units [89,90], artificial neuromorphic visual systems are implemented with optoelectronic synaptic devices, which integrate the functions of optical sensing, signal processing, and in-memory computing [91]. For instance, a reconfigurable WSe_2 photodiode array was reported in which the synaptic weight was stored by a photoresponsivity matrix [92]. The as-constructed neuromorphic vision sensor achieved ultrafast recognition and encoding of optical images.

Many neuromorphic vision devices are based on heterostructures [93]. A fully solution-processed $(PEA)_2SnI_4$/organic Y6 heterojunction synapse was found to have an absorption spectrum from visible to NIR (Figs. 7a–c) [94]. The phototransistor demonstrated a high responsivity of 10^4 A W^{-1} under visible light and 200 A W^{-1} under NIR light. The EPSC behaviors of biological synapses were imitated by controlling photogenerated

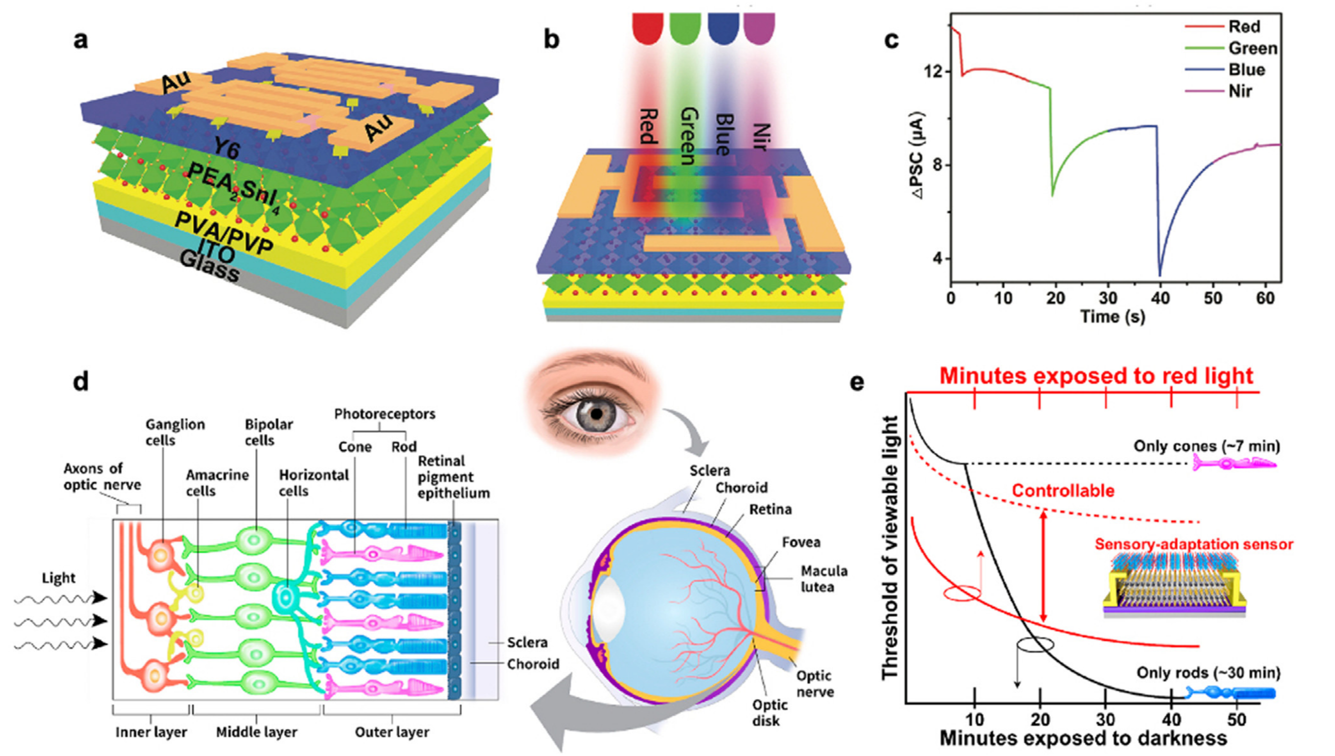

Fig. 7. Demonstration of neuromorphic visual sensor. (a) Schematic illustration of $(PEA)_2SnI_4$/Y6 synapse. (b) Synaptic transistors under stimuli of four light beams. (c) ΔPSC triggered by red, green, blue, and NIR light with the same light power. (a–c) Reproduced with permission. from Ref. [94]. Copyright 2021, Wiley-VCH. (d) Scheme of the eye, with a photoreceptor composed of cones and rods. (e) Adaptation curves of the human eye and the $CsPb(Br_{0.5}I_{0.5})_3$−MoS_2 phototransistor. (d, e) Reproduced with permission. from Ref. [96]. Copyright 2020, American Chemical Society.

hole–electron pairs via Sn vacancies at the junction interface. PPF was also realized by exploiting strong photogating effects, and STP was converted to LTP by changing the frequency and number of light spikes.

A multifunctional device composed of a MoS_2/poly (1,3,5-trimethyl-1,3,5-trivinyl cyclo-trisiloxane) heterostructure [95] yielded photon-triggered synaptic plasticity and realized both image acquisition and data pre-processing functions. Hybrid structures of $CsPb(Br_{1-x}I_x)_3$ perovskite nanocrystals and MoS_2 as phototransistors mimicked artificial synaptic functioning (Figs. 6d and e) [96]. With largely enhanced light absorption induced by the perovskite layer, photocurrents of MoS_2 FETs were improved under visible light owing to charge-carrier transfer between the hybrid layers. Under red light, the currents significantly decreased by two orders of magnitude, as halide segregation of $CsPb(Br_{1-x}I_x)_3$ widened the energy bandgap and thereby degraded the photocurrents. When the light source was removed, the photocurrents recovered to their original values. This time-resolved phenomenon was used to simulate sensory adaptation behaviors.

4.2. Optical neuromorphic computing

It is well accepted that in relation to data computation, the von Neumann architecture limits the data shuttle rate and transistor density. Therefore, in-memory computing and transistor-based computing are considered the next steps, since the former assembles a memory module in a crossbar array and eliminates energy and time consumption, while the latter reduces the transistor feature size and improves the device density [97–99].

Memory cell performance is essential for solving perception tasks, and 2D memory devices such as memristors and memtransistors have low working voltages and reduced power consumption due to their ultrathin layered structures [100]. 2D transistors facilitate large-scale integration for the further fusion of memory and computing, and the number of necessary transistors can be greatly reduced compared with traditional logic computing gates [101]. TMDs with a bandgap of 1–2 eV are appealing for their strong light absorption and illustrate the balance between carrier mobility and current on/off ratio.

For ANNs, high recognition accuracy requires linear and symmetrical weight characteristics, multiple conductive states, and non-volatile memory. An interfacial co-assembly strategy has been used to synthesize heterostructures with perylene as the semiconducting building block layer and GO to increase light absorption in the NIR to UV region (Figs. 8a–c) [102]. The force of hydrogen bond interactions and π–π electron interactions endowed the heterolayers with broadband light absorption (365–1550 nm) and a strong light–matter coupling effect. The heterostructure had a Type II band structure, and the photogenerated electrons were trapped within oxygen-induced defects. The slow release of trapped carriers provided optoelectronic synaptic plasticity, including EPSC, PPF, STP, LTP, spike-intensity dependent plasticity (SIDP), and spike-number dependent plasticity (SNDP), as well as a PPF index as high as 214%. When irradiated by 365 nm pulses at different intervals, the photonic synapse fit Ebbinghaus's learning mechanism. In addition, the high PPF characteristics were exploited by constructing a light-sensitive ANN to mimic a visual learning procedure.

Graphdiyne is an emerging 2D carbon material with natural semiconducting properties. A fully optical two-terminal synapse in a pyrenyl graphdiyne/graphene/PbS heterostructure was reported (Figs. 8d–f) [103]. The aligned energy band structure endowed the hybrids with both negative and positive persistent photoconductivity under different illuminations, and this provided a framework for synaptic plasticity emulation in optical ways. The synapses were used in a multilayer ANN to recognize digits, achieving high accuracy and fault tolerance. A real-time image sensor and logic gates were also realized.

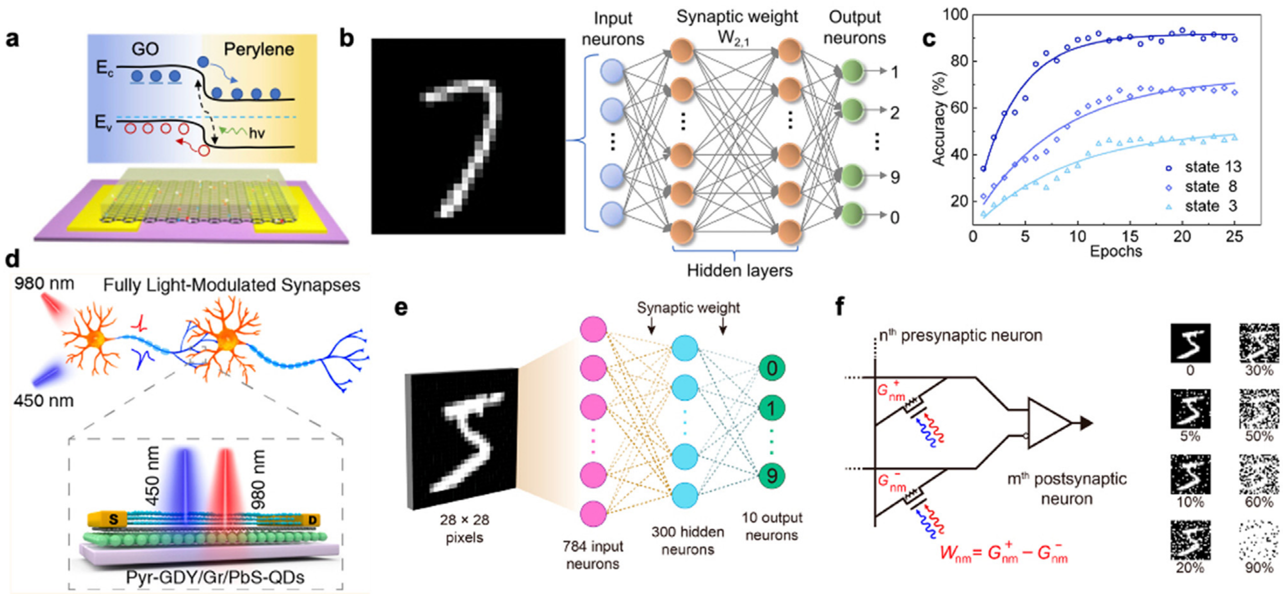

Fig. 8. Illustration of neuromorphic computing system. (a) Scheme and energy band diagram of the perylene/GO heterostructure device. (b) Schematic illustration of a light-driven neural network. (c) Accuracy curves of the designed network under different pulses. (a–c) Reproduced with permission. from Ref. [102]. Copyright 2022, Springer Nature. (d) Schematic of a three-layer neural network with 28 × 28 pixels for handwritten digit recognition. (e) Illustration of the synaptic weight, defined as the conductance difference between two optical synapses. (f) Handwritten digits with different noise ratios. (d–f) Reproduced with permission. from Ref. [103]. Copyright 2021, American Chemical Society.

5. Summary and outlook

The use of diverse 2D materials has provided great flexibility in the design and preparation of synaptic devices, and 2D heterostructures composed of overlapping 2D materials have received extensive attention. Optical synapses are potential substitutes for electrical synapses owing to their large bandwidth, fast processing speed, and low power consumption. This review has presented the basic materials and device responses of representative 2D artificial OEASs. Examples of typical 2D OEASs that have shown various neuromorphic features were summarized, with a focus on the emerging trends in this intensely thriving application area. We described a number of enhancements, with a focus on: (i) gaining an understanding of photocarrier dynamics in 2D materials by modulating their properties; (ii) feasibly engineering surface defects and interfaces by optimizing synthesis processes; and (iii) integrating well-designed device structures and 2D OEASs for neuromorphic applications.

In terms of synthesizing 2D materials and heterostructures, the performance of 2D optoelectronics is quite sensitive to defects and extrinsic disorders arising from the atomically thin flakes that are involved. Defect engineering plays a prominent role in modulating synaptic characteristics, for it has a strong influence on the charge trapping/de-trapping process and synaptic behaviors. The transfer methods to fabricate 2D heterostructures inevitably introduce unwanted wrinkles, defects, and interface contamination, causing device damage and performance degradation. Thus, a tailored synthesis process is still needed. CVD is an alternative strategy to controllably fabricate 2D heterostructures with few defects and high homogeneity. However, the CVD process entails complex preparation and offers low controllability, and the contamination introduced by various precursors greatly affects the composition and quality of 2D materials [104]. In the future, the development of 2D heterostructures will focus on eliminating the influence of cross-contamination, by such means as modulating the growth temperature, controlling precursor vapors, and designing alloy substrates. It is also important to improve carrier mobility through controlling grain sizes and grain boundaries, as this will help optimize the optoelectronic performance of 2D devices.

In terms of realizing large-scale, stable 2D optoelectronics, it is quite challenging to produce high-quality, large-area single crystalline 2D films. CVD has already produced crystalline films of several 2D materials, but a universal method to produce most 2D materials is still being sought. To solve this problem, the growth mechanism of 2D materials should be explored theoretically. Furthermore, methods to control nucleation density and find suitable substrates for TMDs remain to be discovered. Thus, producing large amounts of 2D materials and heterostructures with target properties requires in-depth research into their chemistry and growth mechanisms.

With respect to 2D optoelectronic synaptic devices, there is still a long way to go in achieving OEASs with higher operating speeds and lower energy consumption. The probable solutions include modifications of material properties, well-designed device structures, and exploration of physical mechanisms. The photoresponse of optoelectronic synaptic transistors is largely limited by the contact conditions between the source/drain electrodes and the semiconductor channels, presenting one of the main problems affecting the performance of 2D transistors. For future applications, 2D synaptic devices with multiple terminals are desired to achieve complex functions, and ultrafast neuromorphic applications might be achieved through developing fully optical hardware synapses [109]. In addition, a transistor-structured device that employs the tunneling effects of transport carriers is one of the most promising strategies to achieve low-power OASs. To improve the optoelectronic currents and reduce the operating speed of devices, the Schottky barriers must be adjusted by such methods as interface engineering, surface doping, and phase-change engineering. Moreover, due to the dangling-bond-free surfaces of 2D materials which avoid interface lattice mismatch, constructing 2D heterostructure-transistors that have ultrafast response would reduce carrier thermal emissions and yield high-speed, low-power OEASs.

When it comes to 2D integration, it is important to incorporate multifunctional synaptic functions into 2D neuromorphic systems. High energy efficiency and area efficiency will be the ultimate goals of next-generation neuromorphic devices. Due to the reduced power consumption and ongoing downscaling of novel 2D architectures, integrating 2D materials into a system for visual sensing and computing will offer new possibilities for optimizing energy efficiency [105–108]. Moreover, 2D materials allow high-density integration with a dangling-bond-free lattice, significantly increasing the system area efficiency and thus improving computational performance. The integration of 2D OEASs and silicon circuits is also under investigation [110].

In brief, continued progress at the chip level still requires significant cooperative efforts in the areas of material synthesis, device fabrication, and system architecture design. The lab-to-fab transition is bound to expand in both breadth and depth, given the continued interest in and dedication to 2D optoelectronics.

Author contributions

W. Xu initiated and supervised the research. J. Chen and W. Xu proposed the concept. J. Chen wrote the manuscript, and W. Xu revised the manuscript. All the authors participated in discussions on the manuscript.

Declaration of competing interest

The authors declare no competing financial interests.

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